

Struct

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definition: a collection of **heterogeneous** objects(diff kinds)

each object is called member(array homogeneous同种的)

members' datatype can be different->data type id either basic data type or user defined(enum) or a pointer to the new struct currently being defined

```
struct struct-identifier{  
    data-type identifier-for-1st-member;  
}
```

data type can be basic/userdefined/**a pointer to the new struct**
the struct definition just defines a new user defined data type not an object. usually defined globally.

struct1.x----->way to access x in struct1

```
struct Example {  
    int x;  
    float y[5];  
};  
  
Example a, b;  
b = a; // 这是合法的结构体整体赋值，会复制所有成员包括数组
```

3. 图片还展示了与之不同的逐个成员赋值方式:

```
c  
  
b.x = a.x; // 这样赋值单个成员是可以的  
b.y = a.y; // 错误！数组不能直接这样赋值
```

struct-struct assignment: memberwise copy(member by member, even array can be copied)

separate memberwise assignments: eg p1.x=p2.x ->cannot copy array!

struct initialization:

1. using braces Student_Record a={"Adam", 12345, 'M', CSE, { 2006, 9, 1}}

2. using function

```
void init_date(Date& x,  
    unsigned int year,  
    unsigned int month,  
    unsigned int day)  
{  
    x.year = year;  
    x.month = month;  
    x.day = day;  
}
```

array of structures

```
Student_Record sr[] = {  
    { "Adam", 12000, 'M', CSE, { 2006, 1, 10 } },  
    { "Bob", 11000, 'M', MATH, { 2005, 9, 1 } },  
    { "Cathy", 10000, 'F', ECE, { 2006, 8, 20 } }  
};
```